10/8/24

Practical-5

Experiment on Packet Capture Tool: Wireshark

Aim:
To capture, filter and analyze network packets using the Wireshark packet capture tool.

Description:

Wireshark:

Wireshark, a network analysis tool formerly known as Ethereal, captures packets in real time and display them in human-readable format. Wireshark includes filters, color coding and other features that let you dig deep into network traffic and inspect individual packets. You can use Wireshark to inspect a suspicious program's network traffic, analyze the traffic flow on your network, or troubleshoot network problems.

Capturing Packets:
After downloading and installing Wireshark, launch it and double-click the name of a network interface under Capture to start capturing packets on that interface. As soon as you click the interface's name, you'll see the packets start to appear in real time. Wireshark captures each packet sent to or from your system.

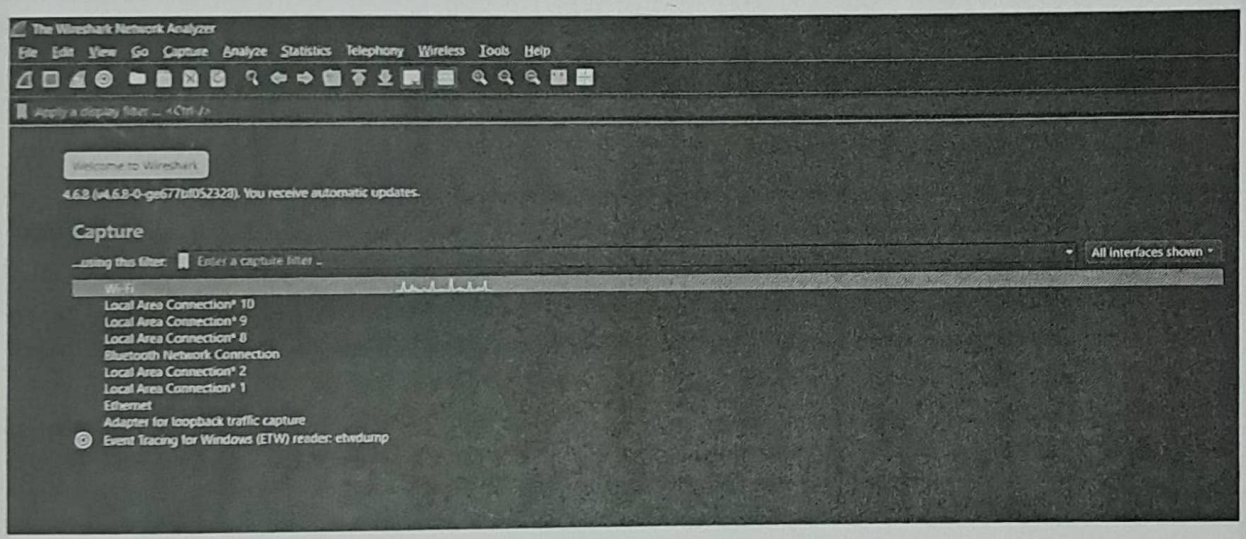
Filtering Packets:
If you're trying to inspect something specific, such as the traffic a program sends when phoning home, it helps to close down all other applications using the network so you can narrow down the traffic. Still, you'll likely have a large amount of packets to sift through. That's where Wireshark's filter come in. The most basic way to apply a filter is by typing in into the filter box at the top of the

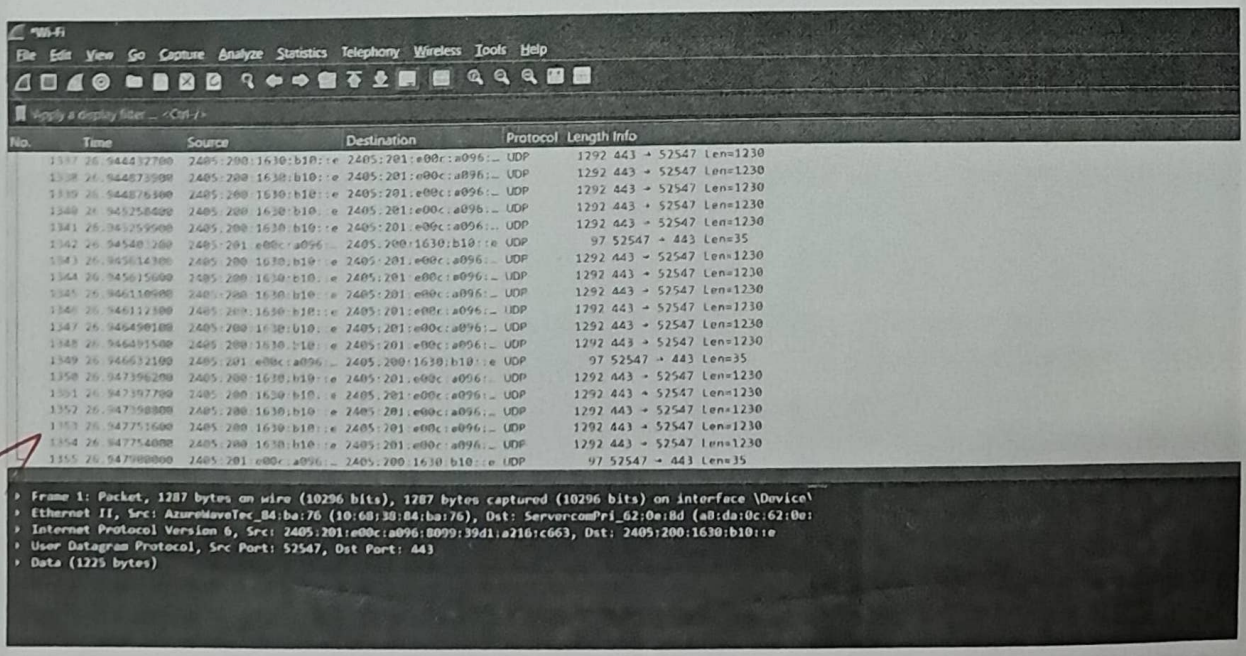
window and clicking Apply. For example, type "dns" and you'll see only DNS packets. When you start typing, Wireshark will help you autocomplete your filter.

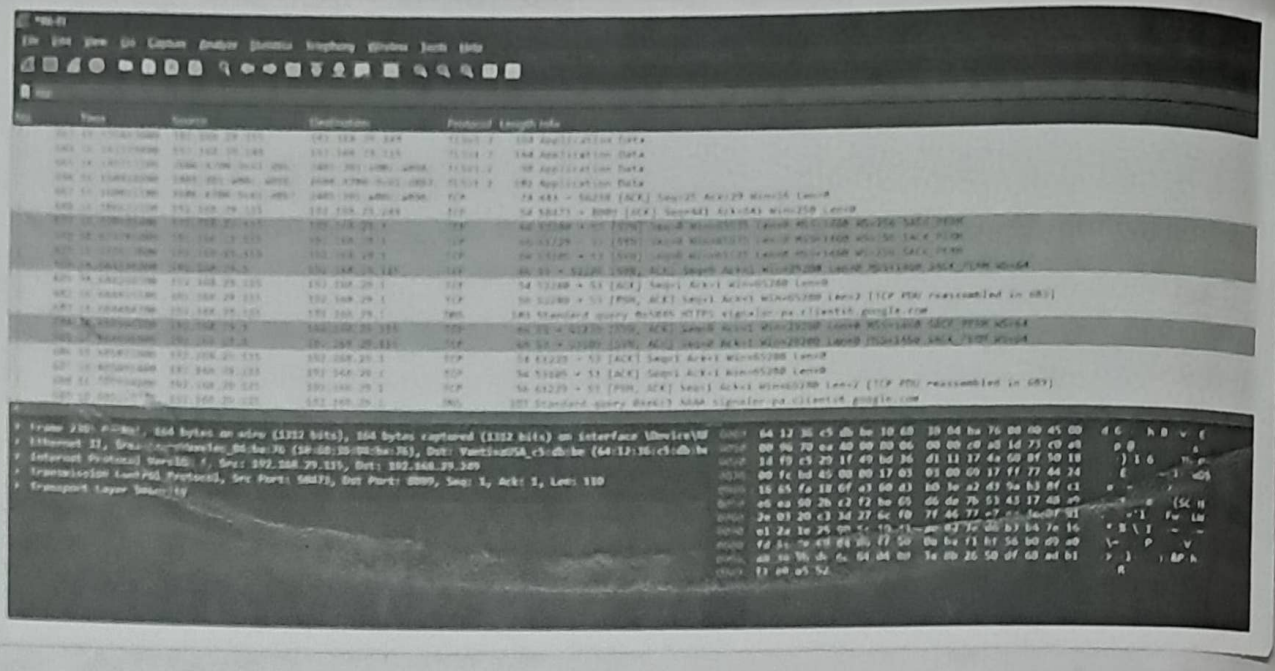
Filtering TCP Packets

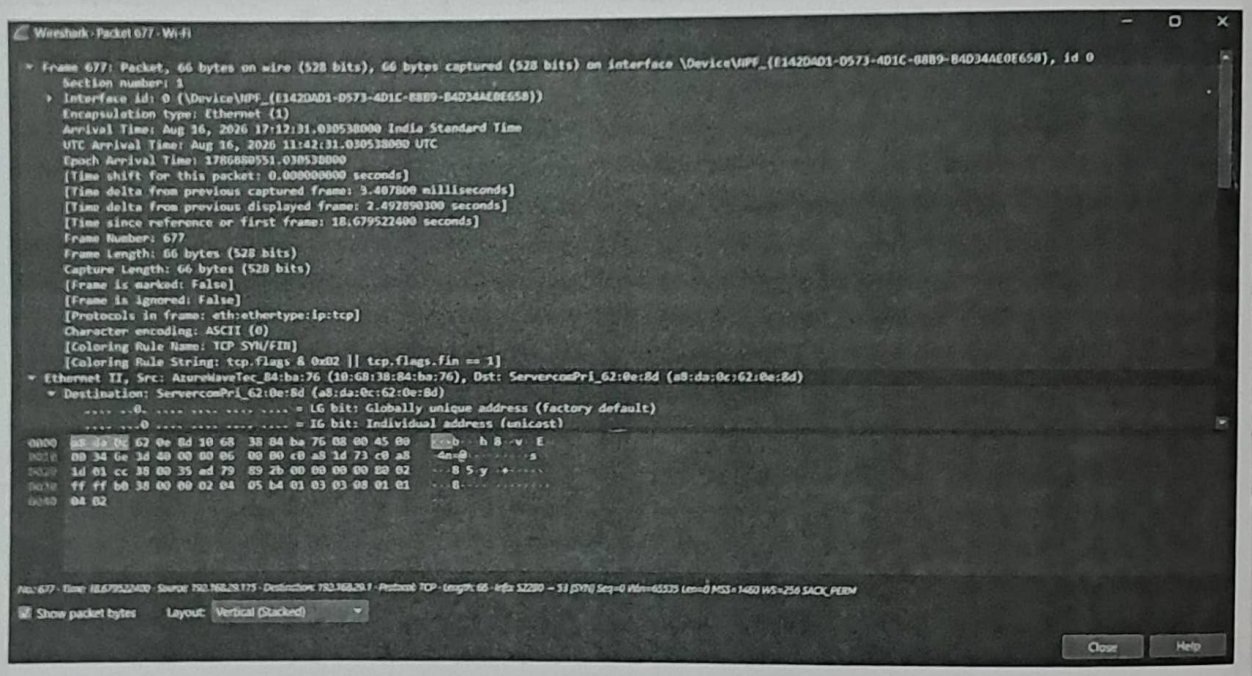
Procedure:

* Select Local Area Connection in Wireshark
* Go to capture option
* Select stop capture automatically after 100 packets
* Then click Start capture
* Search TCP packets in search bar









Result:

Thus the packets are captured, filtered and analysed using the wireshark packet capture tool.